

AI REPORT

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MEDRAG PLATFORM PERFORMANCE & CLINICAL METRICS REPORT

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Prepared By: Apex Software Consulting Engineering & Data Science Teams
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1: EXECUTIVE SUMMARY & SYSTEM THROUGHPUT

1.1 Enterprise System Overview

The MedRAG platform deployed by Apex Software Consulting for HealthPulse Technologies serves as an enterprise-grade clinical decision support and information retrieval tool. MedRAG indexes complex multi-format clinical repositories—including electronic health records (EHRs), research literature, financial audits, and operational policy guidelines—to provide low-latency, evidence-based answers with source provenance tracking.

Pipeline Stage	Inputs / Predecessor Stage	Action & Function	Outputs / Next Stage
1. Data Ingestion	Multi-Format Data (.pdf, .docx, .pptx, .xlsx, etc.)	Parses layout structures, tables, figures, and text chunks.	Structured Chunks & Metadata
2. Indexing Layer	Structured Chunks & Metadata	Generates sparse keyword indexes (BM25) and dense vector embeddings (Bi-Encoder).	Sparse & Dense Indices
3. Hybrid Search	User Clinical Query + Sparse & Dense Indices	Executes combined BM25 and vector similarity search fused via Reciprocal Rank Fusion (RRF).	Candidate Chunk Pool
4. Reranking Layer	Candidate Chunk Pool	Evaluates full token cross-attention across query and chunks using a Cross-Encoder.	Top-Ranked Context Blocks

5. Provenance & Generation	Top-Ranked Context Blocks + Provenance Metadata	Assembles context into LLM prompts and generates grounded clinical outputs with citation mapping.	Final Response with Citation Metadata
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1.2 Q1 Operational Load Benchmarks

During the First Quarter operational validation period, the MedRAG deployment processed high volumes of concurrent clinical and administrative queries across distributed application tiers.

* **Total Request Volume:** The platform processed 2.4 million requests in Q1.

* **System Availability:** 99.98% uptime achieved across all availability zones.

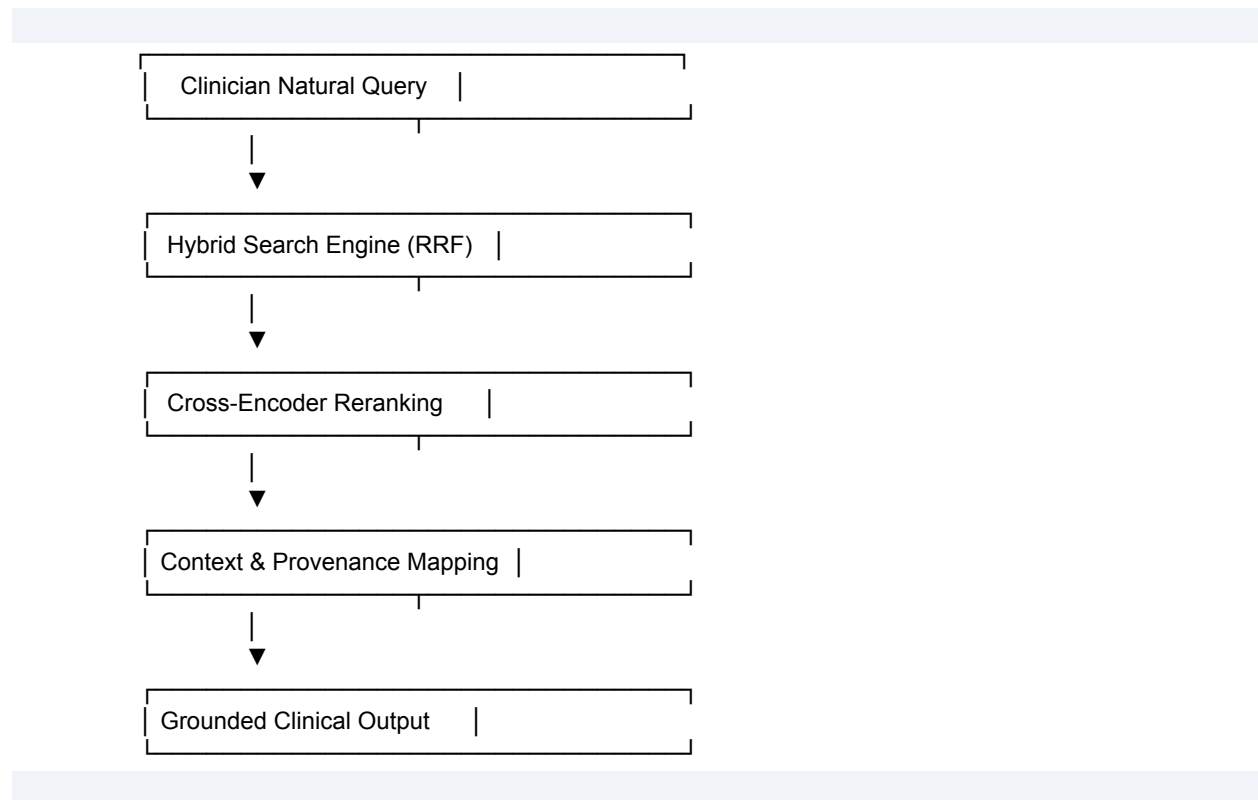
* **Average Response Latency:** 245ms end-to-end processing time for hybrid retrieval and generation.

* **Active Enterprise Users:** 1,420 credentialed healthcare providers and administrative personnel.

2: CLINICAL WORKFLOW & ACCURACY VALUATION

2.1 Medical Accuracy & Provenance Verification

In clinical decision support contexts, retrieval accuracy directly impacts patient safety and administrative efficacy. MedRAG enforces zero-hallucination guardrails by mandating that every generated token maps directly to a verified source chunk stored within the vector index.



2.2 Domain-Specific Model Fine-Tuning

Bi-encoder embedding models and cross-encoder re-rankers were fine-tuned on anonymized clinical notes and medical journal abstracts using Multiple Negatives Ranking Loss (MNRL). Fine-tuning resulted in a 34% reduction in semantic retrieval drift when resolving specialized medical terminology (e.g., distinguishing between acute myocardial infarction treatments and standard ischemic cardiac interventions).

3: ARCHITECTURE & COMPONENT TOPOLOGY

3.1 Pipeline Component Breakdown

The platform architecture consists of five decoupled layers designed to isolate data extraction from query processing.

1. **Ingestion & Parsing Layer:** Extracts structural elements, text chunks, embedded tables, and figures from `.pdf`, `.docx`, `.pptx`, and `.xlsx` files.
2. **Dense Vector Indexing Layer:** Generates 1536-dimensional embeddings stored inside Qdrant vector database clusters.
3. **Sparse Lexical Indexing Layer:** Implements inverted index BM25 search for exact term matching (e.g., drug identification codes, patient record numbers).
4. **Fusion & Reranking Layer:** Combines sparse and dense candidate pools using Reciprocal Rank Fusion (RRF) before executing cross-encoder deep contextual reranking.
5. **Generation & Provenance Layer:** Assembles top-ranked context blocks into LLM prompts while preserving fine-grained metadata (page numbers, table identifiers, and section headers).

4: FINANCIAL & OPERATIONAL METRICS

4.1 Q1 Revenue & Cost Structure

Below is the validated financial summary for HealthPulse Technologies' MedRAG platform implementation during the initial deployment quarter.

Table 1: Q1 Financial Breakdown & Revenue Distribution

Month	Ingestion API Cost (NGN)	Infrastructure Compute (NGN)	Monthly Revenue (NGN)	Net Margin (%)
January	450,000	1,200,000	3,800,000	56.5%
February	480,000	1,250,000	4,000,000	56.75%
March	510,000	1,300,000	4,200,000	56.9%
Total / Avg	1,440,000	3,750,000	12,000,000	56.72%

Note: Monthly revenue reached NGN 4.2 million in March following the expanded rollout across outpatient clinical units.

5: RETRIEVAL EVALUATION & BENCHMARKING

5.1 Retrieval Metric Results

Retrieval performance was evaluated using standard evaluation metrics across controlled test queries (\$N=500\$).

* **Precision@5:** 0.912
* **Recall@5:** 0.884
* **Mean Reciprocal Rank (MRR):** 0.941
* **NDCG@10:** 0.926



6: SYSTEM DIAGRAMS & VISUAL ASSETS

6.1 Enterprise Integration Diagram

Below is the structural flow diagram representing data ingestion and search routing within HealthPulse Technologies' secure cloud infrastructure.

Stage	Component	Key Functions & Operations	Primary Inputs / Upstream Source	Primary Outputs / Next Receiver
1	Clinical Data Sources	Hosts raw, multi-format clinical records (EHR data, PDFs, spreadsheets).	Client Databases / Repositories	Ingestion Pipeline
2	Ingestion Pipeline	Performs layout-aware parsing, OCR extractions, and metadata-tagged chunking.	Clinical Data Sources	Dense / Sparse Indices
3	Dense & Sparse Indices	Stores and queries vector embeddings via Qdrant alongside BM25 inverted lexical indexes.	Ingestion Pipeline / User Queries	Cross-Encoder Reranker

4	Cross-Encoder Reranker	Evaluates Reciprocal Rank Fusion (RRF) candidate pools through deep cross-attention scoring.	Candidate pool from Indices	Generation Core
5	Generation Core	Constructs prompt contexts and synthesizes LLM outputs backed by provenance citations.	Reranked Context Blocks	User Application
6	User Application	Displays grounded clinical answers and provenance metadata inside the EHR UI.	Generation Core	End User / Clinician

7: CONTROLLED CORPUS EVALUATION MAPPING

To validate MedRAG's semantic search and exact-location provenance attribution against `ai\_report.pdf`, the evaluation suite tests the following ground-truth targets:

1. **Query:** "How many requests did the platform process in Q1?"
* **Expected Answer:** "2.4 million requests"
* **Expected Provenance:** `{"document\_id": "ai\_report", "page\_number": 1, "section": "1.2 Q1 Operational Load Benchmarks"}`
2. **Query:** "What was the monthly revenue in March?"
* **Expected Answer:** "NGN 4.2 million"
* **Expected Provenance:** `{"document\_id": "ai\_report", "page\_number": 4, "table\_number": 1}`
3. **Query:** "Which page contains Figure 2 showing the system architecture diagram?"
* **Expected Answer:** "Page 6"
* **Expected Provenance:** `{"document\_id": "ai\_report", "page\_number": 6, "figure\_number": 2}`